Ashley Ann Benoy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.32
GEAR Innovative International School	Class XII (CBSE)	2023	97.20%
GEAR Innovative International School	Class X (CBSE)	2021	99.20%

SKILLS

Language Proficiency: English, French, Japanese, Hindi, Malayalam

Programming Languages: Python, C, C++, Verilog, Bash Script, LaTeX, SQL

Design & Simulation Tools: LTSpice, Solid Edge, Cadence Virtuoso, KiCAD, Altium Designer, OrCAD Capture, PSpice, MATLAB, Simulink

WORK EXPERIENCE

Nutanix :Member of Technical Staff (QA) Intern

- Wrote automated Python test workflows to validate system behavior under failure, recovery, and workload-migration scenarios, improving DR test coverage.
- Validated reliability and resiliency of enterprise-scale distributed storage systems on the Disaster Recovery (DR) Systems QA team.
- Worked on building an AI-assisted code-review tool as a part of the Nutanix Intern Hackathon.

L&T Technology Services: Embedded HW and SW Intern

- Simulated analog circuits such as operational amplifiers and center-tapped transformers using PSpice, analyzing key electrical characteristics across configurations.
- Designed schematics in OrCAD Capture for analog blocks, gaining hands-on experience with industry-standard EDA tools.
- Gained exposure to EMI/ EMC mitigation techniques, analog IC design methodology, and Smith Chart applications through technical discussions with senior engineers.

PROJECTS

Customer Churn Prediction using Ensemble Learning

- Built an end-to-end churn prediction pipeline on a 100,000+ customer telecom dataset using Python, XGBoost, CatBoost, and Scikit-Learn.
- Benchmarked five ML algorithms (XGBoost, CatBoost, Gradient Boosting, Random Forest, Logistic Regression) with feature engineering and missing-value handling.
- AUC of 0.692 via cross-validation and hyperparameter tuning.
- Designed a risk-tiered customer segmentation framework.

CORDIC-Based K-Means Implementation

- Designed a low-complexity K-Means clustering architecture using a Coordinate Rotation Digital Computer (CORDIC) engine to reduce computational overhead in Verilog.
- Generalised the 2D CORDIC-based Euclidean distance computation for n-dimensional vectors, enabling a recursive and hardware-reusable design

RTL Synthesis and Benchmark Analysis

- Synthesized and analyzed RTL designs/benchmarks (ISCAS-85, CEP, ITC99, UART) using the Cadence Genus tool as a part of the Productivity tools in IC design course.

- Compared area, timing, and power metrics across various process corners and operating frequencies.
- Automated design execution and report parsing through TCL scripting, enhancing efficiency and accuracy.

Ping Pong Saga: Python-based Arcade Game

- Developed a two-player Ping Pong game in Python using Object-Oriented Programming (OOP), implementing all core game mechanics like collision detection and scoring with the turtle library.
- Built a graphical user interface (GUI) with tkinter, featuring player name input, a dynamic theming system for custom visuals and audio, and a full menu flow.

Frostbite Detection Socks

- Team project under the Electronics Club Elektronica, initiated from a problem statement of the Indian Armed Forces (IAF).
- Developing smart socks with temperature sensors for real-time frostbite risk detection.
- Implemented a microcontroller-based system to analyze temperature trends and trigger alerts.
- Built for -40°C conditions, with a tested proof-of-concept prototype on a breadboard. Currently working towards building a prototype.

CERTIFICATIONS

GCI World April 2026 : The University of Tokyo (*Matsuo-Iwasawa Laboratory, Graduate School of Engineering*)

- Completed a data science program covering the fundamental theories and implementations of data science.

IBM Databases and SQL for Data Science with Python: Coursera

- Completed training in SQL, relational databases, and data analysis using Python and Pandas.

Python for Engineers and Scientists: Udemy

- Hands-on practice with Sympy, Numpy, Pandas, Matplotlib & Scipy for scientific computing and data handling.

RELEVANT COURSES

Programming: Productivity Tools for IC Design, Numerical Processes for Device Modeling, Introduction to Programming, Embedded Systems

Mathematics: Introduction to Statistics, Introduction to Probability, Probability and Random Variables, Complex Variables,Calculus

POSITIONS OF RESPONSIBILITY

Mentor, Sunshine IIT Hyderabad

- Provided wellness counseling and academic guidance to a cohort of 20 students.
- Organized events for Mental Health Awareness Week 2024, promoting campus-wide mental health advocacy.
- Built strong interpersonal relationships, supporting students through academic and personal challenges

Core Member, Elektronica Club, IIT Hyderabad

- Co-developed frostbite detection socks using temperature sensors and microcontrollers to monitor and alert users to extreme cold exposure.
- Worked on a gesture-controlled robotic arm project using Raspberry Pi 4.
- Organized workshops and interactive sessions to promote interdisciplinary engagement in electronics and signal processing among students from diverse majors.

EXTRACURRICULAR

JENESYS 2025 Scholar: *Top 1% (10 selected / 1,000+ applicants)*

- Selected for a fully funded Japan-India international exchange programme; engaged in institutional visits, workshops, and cross-cultural leadership activities across Japan.

National Sports Organization (NSO), IIT Hyderabad: Athletics

- Chose athletics as my sport of choice, helping me stay active, unwind and recharge after a long day of classes and labs.